

torch.signal.windows.kaiser#

<https://pytorch.org/docs/stable/generated/torch.signal.windows.kaiser.html>

Description:

Computes the Kaiser window. The Kaiser window is defined as follows: where I_0 is the zeroth order modified Bessel function of the first kind (see `torch.special.i0()`), and $N = M - 1$ if `sym` else M . The window is normalized to 1 (maximum value is 1). However, the 1 doesn't appear if M is even and `sym` is `True`. M (int) – the length of the window. In other words, the number of points of the returned window.

Function Signature

```
torch.signal.windows.kaiser(M, *, beta=12.0, sym=True,
dtype=None, layout=torch.strided, device=None,
requires_grad=False)[source]#
```

Parameters

Keyword Arguments

Return type

Parameters

Keyword

beta (float, optional) – shape parameter for the window. Must be non-negative. Default: 12.0 sym (bool, optional) – If False, returns a periodic window suitable for use in spectral analysis. If True, returns a symmetric window suitable for use in filter design. Default: True. dtype (torch.dtype, optional) – the desired data type of returned tensor. Default: if None, uses a global default (see torch.set_default_dtype()). layout (torch.layout, optional) – the desired layout of returned Tensor. Default: torch.strided. device (torch.device, optional) – the desired device of returned tensor. Default: if None, uses the current device for the default tensor type (see torch.set_default_device()). device will be the CPU for CPU tensor types and the current CUDA device for CUDA tensor types. require

Returns:

Tensor

Code Examples

```
>>> # Generates a symmetric gaussian window with a standard
deviation of 1.0.
>>> torch.signal.windows.kaiser(5)
tensor([4.0065e-05, 2.1875e-03, 4.3937e-02, 3.2465e-01, 8.8250e-
01, 8.8250e-01, 3.2465e-01, 4.3937e-02, 2.1875e-03, 4.0065e-05])
>>> # Generates a periodic gaussian window and standard
deviation equal to 0.9.
>>> torch.signal.windows.kaiser(5, sym=False, std=0.9)
tensor([1.9858e-07, 5.1365e-05, 3.8659e-03, 8.4658e-02, 5.3941e-
01, 1.0000e+00, 5.3941e-01, 8.4658e-02, 3.8659e-03, 5.1365e-05])
```

Detailed Documentation

Documentation

Computes the Kaiser window.

The Kaiser window is defined as follows:

where I_0 is the zeroth order modified Bessel function of the first kind (see `torch.special.i0()`), and $N = M - 1$ if `sym` else M .

The window is normalized to 1 (maximum value is 1). However, the 1 doesn't appear if M is even and `sym` is `True`.

M (int) – the length of the window. In other words, the number of points of the returned window.

`beta` (float, optional) – shape parameter for the window. Must be non-negative. Default: 12.0

`sym` (bool, optional) – If False, returns a periodic window suitable for use in spectral analysis. If True, returns a symmetric window suitable for use in filter design. Default: True.

`dtype` (torch.dtype, optional) – the desired data type of returned tensor. Default: if None, uses a global default (see `torch.set_default_dtype()`).

`layout` (torch.layout, optional) – the desired layout of returned Tensor. Default: `torch.strided`.

`device` (torch.device, optional) – the desired device of returned tensor. Default: if None, uses the current device for the default tensor type (see `torch.set_default_device()`). device will be the CPU for CPU tensor types and the current CUDA device for CUDA tensor types.

`requires_grad` (bool, optional) – If autograd should record operations on the returned tensor. Default: False.